THREE EQUIVALENT DESCRIPTIONS OF THE TOPOLOGY INDUCED BY AN INJECTIVE MAP

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Allowed standard prerequisites (editorial).

- The notion of the topology induced by an injective map of a set into a topological space, and the weakest-topology characterisation. Set up in the author's section on submersive maps, immediately before the counted statement. Fixes the three descriptions whose equivalence the statement asserts.

Lemma 0.1. *Let X be a topological space. Let Y be a set and let $f : Y \to X$ be an injective map of sets. The induced topology on Y is the topology characterized by each of the following statements:*

1. *it is the weakest topology on Y such that f is continuous,*
2. *the open subsets of Y are $f^{-1}(U)$ for $U \subset X$ open,*
3. *the closed subsets of Y are the sets $f^{-1}(Z)$ for $Z \subset X$ closed.*

Proof. The set $\mathcal{T} = \{f^{-1}(U) | U \subset X \text{ open}\}$ is a topology on Y. Firstly, $\emptyset = f^{-1}(\emptyset)$ and $f^{-1}(X) = Y$. So $\mathcal{T}$ contains $\emptyset$ and Y.

Now let $\{V_i\}_{i \in I}$ be a collection of open subsets where $V_i \in \mathcal{T}$ and write $V_i = f^{-1}(U_i)$ where U_i is an open subset of X, then

$$\bigcup\nolimits_{i \in I} V_i = \bigcup\nolimits_{i \in I} f^{-1}(U_i) = f^{-1}\left(\bigcup\nolimits_{i \in I} U_i\right)$$

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So $\bigcup_{i \in I} V_i \in \mathcal{T}$ as $\bigcup_{i \in I} U_i$ is open in X. Now let $V_1, V_2 \in \mathcal{T}$. We have U_1, U_2 open in X such that $V_1 = f^{-1}(U_1)$ and $V_2 = f^{-1}(U_2)$. Then

$$V_1 \cap V_2 = f^{-1}(U_1) \cap f^{-1}(U_2) = f^{-1}(U_1 \cap U_2)$$

So $V_1 \cap V_2 \in \mathcal{T}$ because $U_1 \cap U_2$ is open in X.

Any topology on Y such that f is continuous contains $\mathcal{T}$ according to the definition of a continuous map. Thus $\mathcal{T}$ is indeed the weakest topology on Y such that f is continuous. This proves that (1) and (2) are equivalent.

The equivalence of (2) and (3) follows from the equality $f^{-1}(X \setminus E) = Y \setminus f^{-1}(E)$ for all subsets $E \subset X$. $\square$